



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	8.73 / 10
2022	CBSE, Class XII	Delhi Public School, Kalyanpur Kanpur	94.20%
2020	CBSE, Class X	Delhi Public School, Kalyanpur Kanpur	96.60%

WORK EXPERIENCES

System Architecture and Platform Design | Tech Head | Spring Fest [Jul' 24 - Apr' 25]

- Scaled **backend** for **100M+ API requests** by deploying services with **Nginx**, **Redis caching** on **4 AWS EC2 t3.large** instances
- Handled **80L+ transactions** by integrating **PhonePe API** and implementing auth using **JWT**, **OAuth2** and **rate-limited endpoints**
- Delivered **99.8% OTP/email delivery** using **MSG91**, **Mailer**, **async retry queues** and active monitoring through **CloudWatch**
- Shipped **React Native app** on **Play Store (1K+ downloads)** with **backend integration** for **event access** and **admin tools**

Mobile and Sync Systems | Software Developer Intern | Paralleldots [Dec' 24 - Jan' 25]

- Improved **Android render speed** by **25%** and cut **cold start** by **~3.2s** via **ViewModel scoping**, **lazy loading**, and UI refactors
- Reduced **upload failures** by **~20%** using **S3** for **resumable uploads** with **Cognito-secured background access** and **retry logic**
- Cut **invalid scans** by **~18%** using a **SQLite-backed local queue** with **WorkManager sync** triggered by **network connectivity**
- Lowered **API load** by **~10%** by **locally caching** task history, enabling **instant access** and reducing backend calls after **reconnect**

PROJECTS

DBMS | KeyStoneDB | Prof. Pabitra Mitra [Mar' 25 - Apr' 25]

- Built a **lightweight RDBMS** in **C++** with an **SQL parser** using **Flex** and **Bison** and **persistent storage** backed by **RocksDB**
- Implemented **modular query execution** supporting **15+ SQL** operations over **LSM-tree storage** with **transaction handling**
- Developed a **custom key-value-based storage layer** using **RocksDB**, ensuring **durability** and **correctness** for SQL operations
- Added **70+ unit tests** with **Google Test** to verify **parsing accuracy**, **transaction consistency** and **safe concurrent execution**

Computer Networks | Message Oriented Transport over UDP | Prof. Sandip Chakraborty [Feb' 25 - Mar' 25]

- Constructed a **message-oriented transport layer** over **UDP** using **POSIX sockets**, and **pthread library** for **thread management**
- Oriented packets by managing boundaries, aggregating **512KB messages** across **10K+ UDP transmissions** with **send/recv tables**
- Enabled **thread-safe I/O** across **5 threads** via **mutexes**, integrating a **lightweight garbage collector** to reclaim stale sockets
- Stress-tested retransmission under **30-80% loss**, validating **1M+ packets** over **simulated unreliable channels**

Low-Latency Systems | Event-Driven Trading System | Self Project [Jan' 25 - Feb' 25]

- Built a **low-latency, event-driven trading system** in **C++17**, **simulating institutional-grade pricing, execution, and risk workflows**
- Simulated **7 Treasury bonds** using **6 multithreaded TCP servers** with **lock-free queues**, **async I/O** achieving **<300ms latency**
- Designed **11 modular services** for **pricing, execution, inquiry** and **PV01-based risk**, using **stream-driven** architecture
- Processed full **trade lifecycle** via **4 input** and **2 output sockets**, a **throttled GUI (300ms)**, and **in-memory** orderbook storage

Computer Organization Laboratory | Single Cycle RISC Processor | Prof. Sarani Bhattacharya [Oct' 24 - Nov' 24]

- Designed a **32-bit RISC processor** capable of **arithmetic, shift, branching and memory instructions**, **optimizing for constants**
- Implemented it with **64 registers**, **4x longer jump range** and **128x longer relative-branch** range, as compared to standard MIPS
- Wrote **test-suites** and a custom **assembler** in Python3 for simulation, demonstrated bubble sort, GCD on **Nexys 4 FPGA** board

AWARDS AND ACHIEVEMENTS

- Earned the title of **Expert** with a rating of **1709** on **Codeforces**, a competitive programming platform (handle: *sumitkumar-17*)
- Reached a rating of **1925** on **LeetCode**, achieving **Knight** rank and placing in the top **3%** globally (handle: *SumitKumar-17*)
- Achieved contest ranks of **414 (Div 2 Round 1035)** and **181 (Div 3 Round 1032)** on **Codeforces**, each among **25,000+** participants
- Secured positions of **645 (Weekly Contest 454)** and **236 (Weekly Contest 455)** on **LeetCode**, among **28,000** contestants in each
- Secured **All India Rank-2038** out of **150k** candidates in **JEE Advanced 2022**, conducted by **IIT Bombay**
- Achieved **All India Rank-1870** out of **1.2 million** candidates in **JEE Main 2022**, organized by the **National Testing Agency**
- Obtained **All India Rank-83** in **WBJEE 2022** conducted by the **West Bengal Joint Entrance Examinations Board**

SKILLS AND EXPERTISE

- **Programming Languages:** C/C++, Rust, Go, Elixir, Python, TypeScript, JavaScript, Bash, Awk, Verilog, SQL
- **Libraries/ Frameworks:** React, Next.js, React Native, Node.js, Express, Pandas, NumPy, GraphQL, Prisma, Django, Flask, NumPy
- **Tools/ Platforms:** Docker, Kubernetes, Redis, MongoDB, Postgres, MySQL, AWS, Azure, Git, Databricks, Linux Kernel, Raspberry Pi

COURSEWORK INFORMATION

Computer Sc: Algorithms-I* | Software Engineering* | Operating System* | Computer Networks* | Database Management Systems* | Programming and Data Structures* | Systems Programming* | Compilers* | Computer Organization and Architecture* | Discrete Structures | Formal Language and Automata Theory | Switching Circuits and Logic Design* | Algorithms-II

AI - ML: Machine Learning | Data Analytics

Mathematics: Probability and Statistics | Linear Algebra and Complex Analysis | Transform Calculus *Marked courses have laboratory component

EXTRA CURRICULAR ACTIVITIES

- Served as the **Unit Leader** for **NSS Unit-4**, coordinated activities, managed volunteers, and received the **Best Volunteer Award**
- Represented **Vidyasagar Hall of Residence** in **Opensoft**, served as **Vice-Captain**, and delivered the final presentation
- Contributed to the **NixOS** open source project by writing patches, refining Nix expressions, and improving system builds